

PAUL T. SUMMERS

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EDUCATION

Stanford University, Stanford California
PhD in Geophysics
Stanford University, Stanford California
B.S in Physics, M.S. in Geophysics

September 2018 - June 2024
GPA: 3.94
September 2010 - June 2014
GPA: 3.81, 3.89

RESEARCH AND PROFESSIONAL EXPERIENCE

Tufts University & Rutgers University & Georgia Institute of Technology
Postdoctoral Researcher

August 2024 - Present
Atlanta, Georgia

Funded through Tufts and Rutgers, physically based at Georgia Tech.
Numerical modeling of ice mélange and interactions with ocean currents and glacial dynamics.
Extending existing numerical modeling packages MITgcm and GLACIOME1D in fortran, python.

Stanford University Department of Geophysics
PhD Candidate

September 2018 - June 2024
Stanford, CA

Physical processes controlling Antarctic Shear margin locations, applied to Thwaites Glacier and other ice streams.
Thermomechanical ice flow modeling and ice sounding radar processing techniques focused on Antarctic shear margins.

Dropbox Inc.
Software Developer

August 2014 - July 2018
San Francisco, CA

Designed, built and tested custom APEX solutions within Salesforce CRM to meet business requirements.

Stanford University Department of Geophysics
Researcher, M.S. Candidate

June 2013 - June 2014
Stanford, CA

Investigated mechanics of pre-explosive harmonic tremor in the 2009 Redoubt Volcano eruption.

PUBLICATIONS

Summers, P.T.; Robel, A.A.; Jackson, R.H.; King, M.D.; Amundson, J.M.; Burton, J.C. "Sudden Collapse of Ice Mélange Under Increasing Submarine Melt at Tidewater Glaciers", (*In Review*), Preprint: <https://doi.org/10.21203/rs.3.rs-9859110/v1>

Hoffman, A.O.[★]; **Summers, P.T.**[★]; Suckale, J.; Christianson, K.; Catania, G.; Conway, H. "Late Holocene Stabilization of Conway Ice Ridge," *The Cryosphere*, (2026), <https://doi.org/10.5194/tc-20-3977-2026>

Hehlen, M.; Young, T.J.; Karplus, M.; Kaip, G.; Alley, K.; Cronin, V.; Alaya-Cortez, S.; Cameron, E.; May, D.; Pretorius, A.; Seldon, Y.; **Summers, P.**; Zak, J.; Martin, C. "High annual variability in velocity and strain rates across Thwaites Glacier's eastern shear margin from satellite and GNSS observations," *Journal of Glaciology*, (*In Review*)

Hoffman, A. O.; **et al.** "Geophysical surveys and instrument incubation program across a glacier grounding zone: Eastwind Glacier's contribution to McMurdo ice shelf mass balance," *Journal of Glaciology*, (*In Review*)

May, D.F.; Schroeder, D.M.; **Summers, P.T.**; Teisberg, T.O.; Broome, A.L.; Bienert, N.L.; Zak, J.; Young, T.J.; Karplus, M.S.; Tulaczyk, S. "Polarimetric Multi-Offset Radio-Echo Sounding with a Radio Frequency-Over-Fiber ApRES System," *Journal of Glaciology*, 2025, <https://doi.org/10.1017/jog.2025.10114>

Summers, P.T.; Jackson, R. H.; Robel, A. A. "Sub-grid Parameterization of Iceberg Drag in a Coupled Iceberg-Ocean Model", *The Cryosphere*, 2025, <https://doi.org/10.5194/tc-19-5135-2025>

Teisberg, T. O.; Schroeder, D. M.; **Summers, P.T.**; Morlighem, M. "Measurement of Englacial Velocity Fields With Interferometric Radio Echo Sounders," *Journal of Geophysical Research: Earth Surface*, 2025, <https://doi.org/10.1029/2025JF008286>

Summers, P.T.; Schroeder, D. M.; May, D. F.; Suckale, J. "Evidence for and against temperate ice in Antarctic shear margins from radar-depth sounding data," *Geophysical Research Letters*, 2024, <https://doi.org/10.1029/2023GL106893>

Summers, P.T.; Elseworth, C.W.; Dow, C.F.; Suckale, J. "Migration of the Shear Margins at Thwaites Glacier: Dependence on Basal Conditions and Testability Against Field Data," *Journal of Geophysical Research: Earth Surface*, 2023, <https://doi.org/10.1029/2022JF006958>

Siegfried, M.; Venturelli, R; **et al.** , SALSA Science Team. "The life and death of a subglacial lake in West Antarctica," *Geology* , 2023, <https://doi.org/10.1130/G50995.1>

Bienert, N.; Schroeder D. M.; **Summers, P.T.** "Bistatic Radar Tomography of Shear Margins: Simulated Temperature and Basal Material Inversions," *IEEE Transactions on Geoscience and Remote Sensing*, 2022, <https://doi.org/10.1109/TGRS.2022.3213047>

Summers, P.T.; Dustin M. Schroeder, Matthew R. Siegfried. "Constraining Ice Sheet Basal Sliding and Horizontal Velocity Profiles Using a Stationary Phase Sensitive Radar Sounder," *IEEE International Geoscience and Remote Sensing Symposium*, 2021, <https://doi.org/10.1109/IGARSS47720.2021.9554535>

AWARDS

Editor's Citation for Excellence in Refereeing	June 2026
American Geophysical Union (AGU) Journal of Advances in Modeling Earth Systems (JAMES)	
Department Citizenship Award	June 2024
Stanford Department of Geophysics	
ARCS Scholar	2022 - 2024
Northern California Chapter of the Achievement Rewards for College Scientists, 2x recipient for total of \$101,000	
Best Graduate Poster	May 2023
<i>Research Review Symposium</i>	<i>Stanford Doerr School of Sustainability</i>
Radar Attenuation Signature of Temperate Antarctic Shear Margins	
Stanford Earth Graduate Student Research Grant	Sept 2021
Grant of \$575 for 2021-22 Antarctic field work.	

TEACHING AND MENTORING

Mentor for Undergraduate Research	September 2024 - Current
Mentor to multiple undergraduate researchers at Georgia Tech and Tufts University. Projects on idealized ice sheet modeling, mélange modeling, and ocean modeling. One student presenting research at AGU 2025.	
Guest Lecturer	September 2024 - December 2024
Lead guest lectures at Tufts University on Glacial Flow for a Cryosphere course and at Georgia Tech on Ice Streams for a Glaciology course.	
JIRP Academic Faculty	July 2024 - 2025
<i>Juneau Icefield Research Program</i>	
Lead lectures, discussions, mentored student lead projects, and in-field instruction for 32 Undergraduate and graduate students	
Mentor Graduate Student	October 2019 - June 2024
Peer Mentor for Stanford 1st year PhD students (1 hours per week)	
Tutor Graduate Student	October 2022 - June 2023
1:1 tutor for masters students in upper level math and engineering courses (2 hours per week)	
SESUR Program Assistant	April 2022 - October 2022
<i>Stanford Doerr School of Sustainability</i>	
Coordinate Stanford Earth Summer Undergraduate Research Program including field trips, weekly seminars, various social events. (10 hours per week)	
Mentor for Undergraduate Research	June 2021 - August 2022
<i>Stanford Department of Geophysics</i>	
Mentored Stanford undergraduates modeling various glaciology projects. One student presented work at AGU 2022. (3 hours per week)	
Teaching Assistant	April 2022 - June 2022
<i>Stanford University Department of Geophysics</i>	
Undergraduate geophysical methods course for imaging and characterizing groundwater systems. Partnership with community decision makers to recharge ground water (12 hours per week)	
Teaching Assistant	January 2019 - April 2019
<i>Stanford University Department of Geophysics</i>	
Undergraduate and Graduate course. Continuum mechanics applied to ice sheets and glaciers, water waves and tsunamis, and volcanoes. (6 hours per week)	

PROFESSIONAL DEVELOPMENT

CIRTL@Stanford Teaching Certificate Program	August 2023
<i>Associate Level</i>	<i>Stanford, CA</i>
Center for the Integration of Research, Teaching, and Learning (CIRTL) Network.	
Outdoor Leadership Apprenticeship	March 2023 - June 2024
<i>Apprentice</i>	<i>Stanford, CA</i>
Apprenticeship in Outdoor Leadership, focused on rock climbing skills through experiential learning. Co-instructed 2x week long, field-based traditional rock climbing courses for 8 students in Joshua Tree National Park, as well as multiple vertical self-rescue clinics on campus.	

Preparing Future Professors*Mentee*

10 week shadowing program gave the opportunity to experience faculty life first-hand at a comprehensive, teaching-focused university or community college.

November 2022 - Jan 2023

*West Valley College, Saratoga, CA***Center for Teaching and Learning Course Design Institute***Student*

6 week summer course on drafting curricula using evidence-based frameworks. Developed framework of glacial dynamics course focused on mass balance methods.

June 2022 - July 2022

*Stanford, CA***FIELD EXPERIENCE****Juneau Icefield Research Program***Academic Faculty*

Led week long research projects with 6-8 students. Worked with RTK GPS survey, tiltmeters, passive seismometers. Lead field trip based lectures and activities.

July 2024-2025

*Taku Glacier, Juneau Icefield***Thwaites Interdisciplinary Margin Evolution***Field Scientist*

Wide offset (up to 4 km) bistatic, polarimetric radar survey using wireless and fiber optic synchronization techniques using modified pRES radar. Assisted with 2-D and 3-D active seismic survey. Surveyed and Deployed seismic nodes with GPS, assisted in active seismic explosive sources. 7 weeks in the deep field in a team of 16 with 2 guides.

Oct 2023 - Feb 2024

*Thwaites Glacier, West Antarctica***Near-Surface Geophysics: Imaging Groundwater Systems***Teaching Assistant*

Co-lead a class of 20 undergraduates to completed a 100 m seismic (hammer and betsy gun), 200 m electrical resistivity tomography, and towed transient electromagnetic survey imaging ground water connectivity in the top 40 meters of the subsurface. Worked with community decision makers to inform development of newly acquired public lands.

May 2022

*Coyote Valley, California***Thwaites Interdisciplinary Margin Evolution***Field Scientist*

Completed a 5 km offset bistatic, polarimetric radar survey. Deployed and recovered seismic nodes in an active seismic survey using hammer source. Recovered passive seismic nodes and GPS stations. 3 weeks in the deep field in a team of 4 scientists and 2 guides.

Oct 2021 - Jan 2022

*Thwaites Glacier, West Antarctica***TECHNICAL STRENGTHS**

Computer Languages	Python, MATLAB, FORTRAN, Bash, JAVA, SQL, APEX, SOQL, Javascript
Tools	HPC, Git, MITgcm, vim, L ^A T _E X, Sublime IDE
Field Skills	ApRES, Seismic Surveying (Ice and Land), ERT, GPS, Digging in Snow, Roped Travel, Snowmobiling, Crevasse Rescue, Vertical Rock Rescue
First Aid	Red Cross AED, CPR, Basic First Aid Certified (lapsed 2024), WFA (lapsed 2020)

OPEN SOURCE CODE REPOSITORIES**Zenodo**

<https://zenodo.org/records/14721713>
<https://zenodo.org/records/15116445>
<https://zenodo.org/records/10783426>
<https://zenodo.org/record/7106136>

For Publications

Model developed for (Summers, et al. 2025)
 Data for (Summers, et al. 2025)
 (Summers, et. al. 2024)
 (Summers, et. al. 2023)

Github

<https://github.com/somonesummers>
<https://github.com/somonesummers/demoMITgcm>

Ongoing research and personal projects
 Idealizde demo for running MITgcm and iceberg2 package

COMMUNITY BUILDING**Graduate Student Advisory Council Member**

Liaison between graduate students in the Stanford School of Earth and department and school level administration. (1 hour per week)

2019 - 2020

School of Earth Social Czar

Host weekly social events for the Stanford School of Earth. (2 hours per week)

2018 - 2019

CONFERENCE ABSTRACTS**AGU 2025**

Summers, P. T.; Robel, A. A.; Jackson, R. H. (2025, Dec). Rapid Mélange Collapse and De-Buttressing in a Coupled Glacier/Ocean/Mélange Model (GLACIOME)

Dec, 2025

AGU 2025

Dec, 2025

Baker, J.; Robel, A. A.; **Summers, P. T.** (2025, Dec). Quantifying the Feedback Between Ice Mélange Buttressing Force and Glacier Calving.

WAIS 2025 Sept, 2025

Summers, P. T.; Robel, A. A.; Jackson, R. H. (2025, Dec). Glacier-Mélange Feedbacks as a Stabilization Mechanism for Classical Marine Ice Sheet Stability.

Kavli Institute for Theoretical Physics: The Future of Earth's Polar Regions Jun, 2025

Summers, P. T.; Jackson, R. H.; Robel, A. A. (2025, Jun). GLACIOME: The Coupled Glacier/Ocean/Mélange Model. <https://online.kitp.ucsb.edu/online/polar-c25/>

AGU 2024 Dec, 2024

Summers, P. T.; Schroeder, D. M.; May, D. F.; Suckale, J. (2024, Dec). Constraints on the Thermal State of Antarctic Shear Margins from Integration of Thermodynamic Modeling and Airborne Ice Penetrating Radar Data.

AGU 2024 Dec, 2024

May, D. F.; Schroeder, D. M.; **Summers, P. T.**; Teisberg, T. (2024, Dec). Multi-Offset Fiber Optic-Based Radar Arrays For Time-Lapse Imaging of Englacial and Subglacial Processes.

AGU 2024 Dec, 2024

Teisberg, T.; Schroeder, D. M.; **Summers, P. T.**; Morlighem, M. (2024, Dec). Inferring Englacial Velocity from Interferometric Ice-Penetrating Radar Sounding: Opportunities and Challenges in Regions with Complex Internal Dynamics.

WAIS 2024 Nov, 2024

Summers, P. T.; Robel, A. A.; Jackson, R. H. (2024, Dec). Not Your Average Berg: Development of a Coupled Mélange/Ocean Model.

EGU 2024 Apr 18, 2024

Emma C. Smith; **et. al.** (2024, April). Icequakes beneath Thwaites Glacier eastern shear margin.

EGU 2024 Apr 19, 2024

Daniel May; **et. al.** (2024, April). Multi-Offset Radio-Echo Sounding for Estimation of Englacial and Subglacial Thermal Conditions and Material Properties.

WAIS 2023 Meeting Sept 26, 2023

Summers, P. T.; Andrew Hoffman; **et. al.** (2023, May). Historic Shear Margin Migration at Conway Ice Rise: An Integrated Data-Model Approach.

SDSS 2023 Research Review May 26, 2023

Summers, P. T.; Schroeder, D.; Suckale, J. (2023, May). Radar Attenuation Signature of Temperate Antarctic Shear Margins.

AGU 2022 Meeting Dec 13, 2022

Summers, P. T.; **et. al.** (2022, Dec). Response of Thwaites Glacier's Shear Margins to Ice Sheet Thinning and Surface-Slope Steepening. In AGU Fall Meeting Abstracts.

AGU 2022 Meeting Dec 13, 2022

Cheng, C. **et. al.** (2022, Dec). Sensitivity of Subglacial Streams to Bed Topography: Introducing Small-Scale Bed Roughness Suggests Large Water Routing Uncertainties for Thwaites Glacier. In AGU Fall Meeting Abstracts.

AGU 2022 Meeting Dec 13, 2022

Teisberg, T. **et. al.** (2022, Dec). Methods for Constraining Englacial Velocity Fields using Airborne Ice-penetrating Radar Data. In AGU Fall Meeting Abstracts.

WAIS 2022 Meeting Sep 27, 2022

Summers, P. T.; Schroeder, D. (2022, Sep). Evidence for Temperate Ice in Shear Margins of Antarctic Ice Streams from Airborne Radar Surveys.

AGU 2021 Meeting Dec 14, 2021

Siegfried, M. R.; **et. al.** (2021, Dec). The life and death of a subglacial lake in West Antarctica. In AGU Fall Meeting Abstracts.

AGU 2021 Meeting Dec 14, 2021

Sandra, R.; **et. al.** (2021, Dec). Informing Bistatic Radar Experiments at Thwaites Glacier Using Bistatic Data from Greenland and West Antarctica. In AGU Fall Meeting Abstracts.

WAIS Workshop 2021 Sep 22, 2021

Summers, P.T.; Elseworth, C.W.; Suckale, J.; TIME Science Team (2021, Sep). Inward Migration of the Shear Margins at Thwaites Glacier in Response to Thinning.

WAIS Workshop 2021 Sep 23, 2021

Summers, P.T.; Schroeder, D.; Suckale, J. (2021, Sep). Evidence for Temperate Ice in Shear Margins of Antarctic Ice Streams from Airborne Radar Surveys.

IEEE International Geoscience and Remote Sensing Symposium 2021

July 11, 2021

Summers, P.T.; Schroeder, D.; Siegfried, M.R. (2021, July). Constraining Ice Sheet Basal Sliding and Horizontal Velocity Profiles Using A Stationary Phase Sensitive Radar Sounder.

AGU 2020 Meeting

Dec 16, 2020

Summers, P.T.; Elseworth, C.W.; Suckale, J; TIME Science Team (2020, Dec). Processed-Based Models in the Wild: A Forward Model Approach to Constraining the Processes Governing Basal Strength at Thwaites Glacier. In AGU Fall Meeting Abstracts.

WAIS Workshop 2020

Sep 29, 2020

Summers, P.T.; Elseworth, C.W.; Suckale, J; TIME Science Team (2020, Sep). Investigating Mechanisms of Basal Strength at Thwaites Glacier using a Forward Model Approach. Recording of talk on waisworkshop.org

AGU 2019 Meeting

Dec 13, 2019

Summers, P.T.; Elseworth, C.W.; Suckale, J (2019, Dec). Potential Formation of a New Shear Margin at Thwaites Glacier. In AGU Fall Meeting Abstracts.

AGU 2019 Meeting

Dec 13, 2019

Liu, W.; Räss, L.; **Summers, P.**; Papula, A.; Suckale, J. (2019, Dec). Impact of Complex Topography on Thermomechanical Coupled Ice Flow Using the Immersed Boundary Method. In AGU Fall Meeting Abstracts.

SSA 2014 Meeting

May 2, 2014

Summers, P.T. & Dunham, E. M.D. (2014, May). Conduit Processes Driving Pre-explosive Harmonic Tremor in the 2009 Redoubt Volcano Eruption. In SSA 2014 Annual Meeting Announcement.

AGU 2013 Fall Meeting

Dec 2013

Summers, P. & Dunham, E. M. (2013, December). Conduit Processes Driving Pre-explosive Harmonic Tremor in the 2009 Redoubt Volcano Eruption. In AGU Fall Meeting Abstracts.